

二项分布 $X \sim b(n, p)$

$$P(X=k) = C_n^k \cdot p^k \cdot (1-p)^{n-k}, \quad k=0, 1, 2, \dots, n$$

$$E(X) = n \cdot p \quad D(X) = n \cdot p \cdot (1-p) \quad (\text{将 } X \text{ 看成 } n \text{ 个 } 0-1 \text{ 分布的和事件})$$

0-1分布 (特殊的二项分布)

$$E(X) = p, \quad D(X) = p \cdot (1-p)$$

泊松分布 $X \sim \pi(\lambda)$

$$P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k=0, 1, 2, \dots$$

当 $n \geq 20, p \leq \frac{1}{20}$ 时,

$$C_n^k \cdot p^k (1-p)^{n-k} \approx \frac{\lambda^k}{k!} e^{-\lambda}, \quad \text{其中 } \lambda = np.$$

$$E(X) = \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \cdot \lambda \cdot \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda$$

$$D(X) = E(X^2) - [E(X)]^2 = \lambda$$

$$\begin{aligned} E(X^2) &= \sum_{k=0}^{\infty} k^2 \cdot \frac{\lambda^k e^{-\lambda}}{k!} = \sum_{k=1}^{\infty} k^2 \cdot \frac{\lambda^k e^{-\lambda}}{k!} = \lambda \sum_{k=1}^{\infty} (k-1+1) \cdot \frac{\lambda^{k-1} e^{-\lambda}}{(k-1)!} \\ &= \lambda \sum_{k=2}^{\infty} \frac{e^{-\lambda} \lambda^{k-1}}{(k-2)!} + \lambda \cdot \sum_{k=1}^{\infty} \frac{e^{-\lambda} \lambda^{k-1}}{(k-1)!} = \lambda^2 + \lambda \end{aligned}$$

指数分布

$$f(x) = \begin{cases} \frac{1}{\theta} e^{-\frac{1}{\theta}x}, & x > 0 \\ 0, & \text{其他} \end{cases}$$

$$F(x) = \begin{cases} 1 - e^{-\frac{x}{\theta}}, & x > 0 \\ 0, & \text{其他} \end{cases}$$

指数分布无记忆性 即 $P\{X > s+t \mid X > s\} = P\{X > t\}$

$$E(X) = \int_0^{+\infty} x \cdot \frac{1}{\theta} e^{-\frac{1}{\theta}x} dx = \theta$$

$$E(X^2) = \int_0^{+\infty} x^2 \cdot \frac{1}{\theta} e^{-\frac{1}{\theta}x} dx = 2\theta^2$$

$$D(X) = E(X^2) - [E(X)]^2 = \theta^2$$

正态分布 $X \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad E(X) = \mu, \quad D(X) = \sigma^2$$

若 $X \sim N(0, 1)$, 则 $\phi(x) + \phi(-x) = 1$

若 $X \sim N(\mu, \sigma^2)$, 则 $\frac{X-\mu}{\sigma} \sim N(0, 1)$.

随机变量的函数的分布

例: $Y = X^2$

$$F_Y(y) = P\{Y \leq y\} = P\{X^2 \leq y\}$$

$$y > 0 \text{ 时, } F_Y(y) = P\{-\sqrt{y} \leq X \leq \sqrt{y}\} = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

求导可得 $f_Y(y)$.

二维随机变量

边缘分布: $F_X(x) = \lim_{y \rightarrow +\infty} F(x, y) = \int_{-\infty}^x \left[\int_{-\infty}^{+\infty} f(x, y) dy \right] dx$

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$$

$$F_Y(y) = \lim_{x \rightarrow +\infty} F(x, y) = \int_{-\infty}^y \left[\int_{-\infty}^{+\infty} f(x, y) dx \right] dy$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$$

条件分布: $P\{X = x_i | Y = y_j\} = \frac{P\{X = x_i, Y = y_j\}}{P\{Y = y_j\}}$

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

$$F_{X|Y}(x|y) = \int_{-\infty}^x f_{X|Y}(x|y) dx = \int_{-\infty}^x \frac{f(x, y)}{f_Y(y)} dx$$

独立性: $\forall x, y \quad P\{X = x_i, Y = y_j\} = P\{X = x_i\} \cdot P\{Y = y_j\}$

$$F(x, y) = F_X(x) \cdot F_Y(y)$$

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

两个随机变量的函数的分布

设 $(X, Y) \sim f(x, y)$, $Z = g(X, Y) \xrightarrow{\text{解出}} Y = h(x, z)$

则: $f_Z(z) = \int_{-\infty}^{+\infty} f(x, h(x, z)) \left| \frac{\partial h}{\partial z} \right| dx$

$Z = X + Y$ 若 X, Y 相互独立 $\Rightarrow f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) \cdot f_Y(z-x) dx$ (卷积公式)

若 $X \sim N(\mu_1, \sigma_1^2)$ $Y \sim N(\mu_2, \sigma_2^2)$ 且 X, Y 独立 $\Rightarrow Z = X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

$$f_{Y|X}(z) = \int_{-\infty}^{+\infty} |x| \cdot f(x, xz) dx$$

$$f_{XY}(z) = \int_{-\infty}^{+\infty} \frac{1}{|x|} \cdot f(x, \frac{z}{x}) dx$$

$M = \max\{X, Y\} \quad F_{\max}(z) = P\{X \leq z, Y \leq z\} = F_X(z) \cdot F_Y(z)$

$N = \min\{X, Y\} \quad F_{\min}(z) = 1 - P\{X \geq z, Y \geq z\}$

X, Y 相互独立 $= 1 - [(1 - F_X(z)) \cdot (1 - F_Y(z))]$

切比雪夫不等式: $E(X) = \mu, D(X) = \sigma^2$

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$$

$$\text{Cov}(X, Y) = E(X, Y) - E(X) \cdot E(Y)$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{D(X)} \cdot \sqrt{D(Y)}} \quad D(X+Y) = D(X) + D(Y) + 2 \text{Cov}(X, Y)$$

$$\text{Cov}(aX, bY) = ab \cdot \text{Cov}(X, Y)$$

$$\text{Cov}(X_1 + X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$$

$$\rho_{XY} = 0 \Leftrightarrow X, Y \text{ 不相关} \Leftrightarrow \text{Cov}(X, Y) = 0 \Leftrightarrow E(XY) = E(X) \cdot E(Y)$$

(X, Y) 服从二维正态分布时: X, Y 相互独立 $\Leftrightarrow X, Y$ 不相关.

$$(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho) \quad \rho_{XY} = \rho$$

$E(X^k)$ k 阶矩

$E\{[X - E(X)]^k\}$ k 阶中心矩

$E(X^k Y^l)$ $l+k$ 阶混合矩

$E\{[X - E(X)]^k \cdot [Y - E(Y)]^l\}$ $k+l$ 阶中心矩

二维正态分布协方差矩阵

$$C = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

$$f(x_1, x_2) = \frac{1}{(2\pi) \cdot |C|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(X-\mu)^T C^{-1}(X-\mu)\right\}$$

$$f(x_1, x_2) = \frac{1}{2\pi \cdot \sigma_1 \sigma_2 \sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho \frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1 \sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} \right]\right\}$$